Experiment 10

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1. Aim of the Session:

To understand and apply transaction control in PostgreSQL using BEGIN, COMMIT, ROLLBACK, and SAVEPOINT to maintain data integrity during database operations.

2. Software Requirements:

- PostgreSQL
- pgAdmin
- Windows Operating System

3. Objectives of the Session:

Apply transaction control commands for safe and reliable database updates.

4. Theory:

Transaction control in PostgreSQL ensures that a group of database operations is executed safely as a single unit. Using **BEGIN**, a transaction starts; **COMMIT** permanently saves the changes, while **ROLLBACK** undoes them if an error occurs. **SAVEPOINT** allows partial rollback to a specific point within a transaction. These commands help maintain data integrity and consistency, especially during complex or multi-step operations.

5. Practical / Experiment Steps:

Step 0: Table creation and data insertion.

Code:

```
-- Create table
CREATE TABLE payroll (
    emp_id INT PRIMARY KEY,
```

```
    emp_name VARCHAR(50),
    salary DECIMAL(10,2) CHECK (salary > 0)
);

INSERT INTO payroll VALUES
(1, 'Sanju', 30000),
(2, 'Riya', 40000),
(3, 'Sidhi', 50000);

START TRANSACTION;

UPDATE payroll
SET salary = 1000
WHERE emp_id = 3;

UPDATE payroll
SET salary = 1000
WHERE emp_id = 3;

ROLLBACK;

SELECT * FROM payroll;

START TRANSACTION;

UPDATE payroll
SET salary = salary + 5000
WHERE emp_id = 1;

SAVEPOINT sp1;

UPDATE payroll
SET salary = salary + 7000
WHERE emp_id = 2;

UPDATE payroll
SET salary = -1000
WHERE emp_id = 3;

ROLLBACK TO sp1;

COMMIT;
```

SELECT * FROM payroll;

Output:

	emp_id [PK] integer	emp_name character varying (50)	salary numeric (10,2)
1	1	Sanju	30000.00
2	2	Riya	40000.00
3	3	Sidhi	50000.00

	emp_id [PK] integer	emp_name character varying (50)	salary numeric (10,2)
1	2	Riya	40000.00
2	3	Sidhi	50000.00
3	1	Sanju	35000.00

6. Learning Outcomes (What I have learned):

- **BEGIN** starts a transaction block for grouping multiple SQL commands.
- **ROLLBACK** cancels changes made inside the transaction.
- **SAVEPOINT** allows partial rollback without losing all valid updates.
- **CHECK** constraint ensures salary remains greater than zero.
- **PostgreSQL** transaction control helps maintain consistency and data integrity.